

FINANCIAL LITERACY

PATHWAYS TO ARTS & HUMANITIES SUMMER SCHOLARS PROGRAM

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QUIZ: QUESTION 1 OF 4

Question

Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?

- ☐ (a) More than \$102
- ☐ (b) Exactly \$102
- ☐ (c) Less than \$102
- ☐ (d) Do not know

QUIZ: QUESTION 2 OF 4

Question

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?

- ☐ (a) More than today
- ☐ (b) Exactly the same
- ☐ (c) Less than today
- ☐ (d) Do not know

QUIZ: QUESTION 3 OF 4

Question

Please tell me whether this statement is true or false: “Buying a single company’s stock usually provides a safer return than a stock mutual fund.”

- ☐ (a) True
- ☐ (b) False
- ☐ (c) Do not know

QUIZ: QUESTION 4 OF 4

Question

If interest rates rise, what should happen to bond prices?

- ☐ (a) They should rise
- ☐ (b) They should fall
- ☐ (c) They should stay the same
- ☐ (d) There is no relationship between bond prices and interest rates
- ☐ (e) Do not know

HOUSEHOLD FINANCE

Household Finance

The study of how people (households) interact with financial instruments and markets—saving, investing, borrowing, and spending.

Today's big idea: people **systematically** make financial mistakes.

Not because they're dumb—but because **our brains aren't wired for financial decisions**.

OUTLINE

1. Saving & Investing

2. Borrowing Smart

3. Spending & Behavior

4. Recap

WHAT IS THE STOCK MARKET?

Stock Market

A marketplace where people buy and sell **shares**—small pieces of ownership in companies like Apple, Google, or Nike.

Why does it matter for you?

- Historically, stocks return about **7–10% per year** on average
- A savings account pays about **0.5–4%**
- Over decades, this difference is *enormous*

\$1,000 invested in stocks at age 18 could grow to \$24,000+ by age 65. The same \$1,000 in a savings account? About \$2,500.

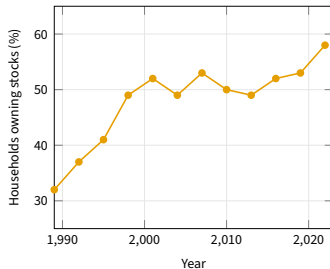
THE PARTICIPATION PUZZLE

The puzzle: If stocks are so good, why doesn't everyone invest?

Facts:

- Only **50–55%** of U.S. households own *any* stocks
- For the bottom half by wealth, participation is only **20–30%**
- In some European countries, it's below **10%**

Economic theory says everyone with positive wealth should hold at least some stock.



Source: Federal Reserve SCF

WHY DON'T PEOPLE INVEST?

1. **Fixed costs** (rational reason)

Learning how, opening accounts, paying fees—about \$300/year

2. **Loss aversion** (behavioral)

“I might lose money”—losses *hurt* about twice as much as gains feel good

3. **Lack of trust** (behavioral)

“The system is rigged”—people who distrust markets avoid them entirely

4. **Low financial literacy** (information)

An extra year of education increases stock market participation by 4 pp

Kahneman & Tversky (1979); Guiso, Sapienza & Zingales (2008); Cole et al. (2014)

YOUR PAST SHAPES YOUR INVESTING

People who **lived through stock market crashes** are less likely to invest—even decades later.

Experience effects:

- If your parents lost money in 2008, your family may avoid stocks today
- People who saw *high* returns in their youth invest more aggressively
- Your generation's financial experiences shape your financial behavior for life

Malmendier & Nagel (2011)



Photo: Pexels (free license)

POLICY SOLUTION: BABY BONDS

The idea: The government opens an investment account for **every child at birth**, invested in low-cost index funds.

How It Works

1. Government deposits \$1,000–\$5,000 at birth
2. Money is invested in a diversified stock index fund
3. Account grows tax-free
4. Accessible at age 18 for education, a home, or starting a business

Why It Helps

- Solves the non-participation puzzle—everyone gets a stake
- Reduces wealth inequality at birth
- Builds financial literacy through ownership
- At 7% return, \$1,000 becomes ~\$3,400 by age 18

ANSWER TO Q1

Question

If you managed to save \$1, how much of it should you put in the stock market?

Answer

Theory says: put *some* in stocks. Even a small amount benefits from the higher average returns. But most Americans don't invest at all—held back by costs, fear of losses, lack of trust, and low financial literacy. Policies like baby bonds could help by giving everyone a stake from birth.

5-minute break

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CREDIT CARDS 101

Credit Card

A card that lets you borrow money from a bank to make purchases. If you don't pay the full balance each month, you owe **interest** on what's left.

Key facts:

- Average interest rate: **20–25% APR**
- Americans have ~**4 cards** on average
- Over half carry a balance month to month



Photo: Pexels (free license)

THE REPAYMENT PUZZLE

**You owe money on three cards. You have \$200 to pay this month.
How do you split it?**

Card A

Department Store
Balance: **\$500**
APR: **22%**

Card B

Bank Card
Balance: **\$300**
APR: **15%**

Card C

Credit Union
Balance: **\$200**
APR: **8%**

Minimum payment: \$25 per card

WHAT PEOPLE ACTUALLY DO

The optimal strategy: Pay minimums on low-rate cards, put the rest on the **highest-rate card**.

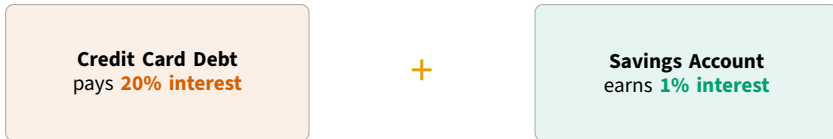
What most people do instead:

- **Balance-matching:** Split payments in proportion to balances—bigger balance gets a bigger payment, *regardless of interest rate*
- **1/N rule:** Split payments equally across all cards (\$67 each)—simple, but costly
- Neither strategy considers which card is *most expensive*

These mistakes are not small. Over a year, the wrong strategy can cost \$20–25 in extra interest—and that adds up fast with larger balances. (Gathergood et al., 2019)

THE CO-HOLDING PUZZLE

Here's an even stranger pattern:



*~12% of households hold **both** at the same time!*

Why not use the savings to pay off the debt?

Explanations: need cash for emergencies, self-control (keeping savings as a “commitment device”), or simply not thinking about it.

Gathergood & Weber (2014); Gross & Souleles (2002)

STUDENT LOANS: WHAT YOU SHOULD KNOW

Key facts for your future:

- Most student loans are **federal**—fixed interest rates, set by Congress
- Repayment starts **6 months after** you leave school
- Unlike credit card debt, student loans are **very difficult to discharge in bankruptcy**—they follow you
- Student debt can affect **career choices** and **homeownership**—harder to qualify for a mortgage

Student loans are an *investment* in yourself. The key is borrowing **wisely**—not more than you need, and understanding the terms *before* you sign.

ANSWER TO Q2

Question

If you owed money on two credit cards, which one should you pay off first?

Answer

Always pay the highest interest rate card first! Pay the minimum on all other cards, then put every extra dollar toward the most expensive one. But research shows most people don't do this—they split payments equally or proportionally by balance, costing themselves real money.

ACTIVITY: THE CREDIT CARD CHALLENGE

Your scenario:

You have three credit cards:

Card	Balance	APR
Card A (Dept. Store)	\$500	22%
Card B (Bank)	\$300	15%
Card C (Credit Union)	\$200	8%

Minimum payment: \$25 per card.

You have **\$200 total** to pay this month.

Your Task

With your group:

How do you split the \$200 across the three cards?

Write your answer on an index card.

THE REVEAL

Strategy	Card A (22%)	Card B (15%)	Card C (8%)	Interest paid over 12 months
Optimal	\$150	\$25	\$25	~ \$85
Balance-matching	\$100	\$60	\$40	~\$105
1/N (equal split)	\$67	\$67	\$67	~ \$110

The 1/N mistake costs you \$25 extra per year—that's real money!

With bigger balances (think: thousands), the cost of the wrong strategy grows to **hundreds of dollars**.

DEBRIEF

What strategy did your group choose?

Why is it hard to focus on interest rates instead of balances?

Think about it: interest rates aren't printed on the front of the card. Balances are on every statement. Our attention is drawn to what's most visible—not what's most important.

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THE PAIN OF PAYING

When you hand over **cash**, your brain registers something like *physical pain*—you feel the money leaving.

When you **swipe a card**? Much less pain.

Result: People spend **more** when paying with cards.

In one experiment, people were willing to pay **twice as much** for basketball tickets when using a credit card vs. cash.

Prelec & Simester (2001)



Photo: Pexels (free license)

MENTAL ACCOUNTING

Mental Accounting

The tendency to treat money differently depending on which “mental bucket” it belongs to—even though a dollar is a dollar.

Examples:

- Retirement savings feel “off limits”—people won’t touch them even when it makes financial sense
- A \$10 birthday gift gets spent freely, but \$10 from your paycheck feels harder to part with
- People assign different credit cards to different “budgets”—ignoring which card has the better rate

Thaler (1999)

PRESENT BIAS

Present Bias

The tendency to prefer a smaller reward **now** over a larger reward **later**—even when waiting is clearly better.

Would you rather have...

A. \$10 today or **B.** \$12 tomorrow?

Most people pick A—even though B is a **20% return overnight**.

Present bias explains why saving is hard: the benefit is in the *future*, but the sacrifice (spending less) is *right now*. Your future self always loses the negotiation.

NUDGES THAT HELP

If our brains work against us, can we design systems that help?

Yes! These are called **nudges**:

- **Automatic enrollment** in retirement savings: participation jumps from ~40% to ~90% just by making it the default
- **Default contribution rates**: People tend to stick with whatever percentage is set for them
- **Credit card statements**: Since 2009 (CARD Act), your statement must show how long it takes to pay off at the minimum—a nudge to pay more

A nudge changes the *default* without removing choice. It helps people do what they *would* want to do if they had perfect self-control. (Madrian & Shea, 2001; Thaler & Sunstein, 2008)

HOW FINANCIALLY LITERATE ARE YOU?

Try the “Big Three” financial literacy questions:

1. Suppose you had \$100 in a savings account at 2% interest per year. After 5 years, how much would you have?
(a) More than \$102 (b) Exactly \$102 (c) Less than \$102
2. If the interest rate on your savings is 1% and inflation is 2%, after one year can you buy more, the same, or less?
(a) More (b) Same (c) Less
3. True or false: buying a single company's stock is usually safer than buying a stock mutual fund.
(a) True (b) False

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(a) True (b) False

Answers: (a), (c), (b) — **Only 43% of U.S. adults get all three right!**

Lusardi & Mitchell (2011); Hastings et al. (2013)

ANSWER TO Q3

Question

Why do you spend more when you pay with a card than with cash?

Answer

Cards remove the “pain of paying.” When you hand over cash, your brain feels the loss. When you swipe a card, that pain signal is muted—so you spend more freely. Add mental accounting (treating different money differently) and present bias (wanting things now), and it’s easy to see why smart people make costly financial mistakes.

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KEY TAKEAWAYS

1. **Investing:** Theory says everyone should own some stocks, but most people don't—costs, fear, and lack of knowledge keep them out. Baby bonds could change that.
2. **Borrowing:** Always pay the highest interest rate first. Most people split payments equally or by balance—a costly mistake.
3. **Spending:** Cards reduce the “pain of paying,” mental accounting distorts our choices, and present bias makes saving hard. Nudges can help.

WHAT CAN YOU DO?

1. Start investing early

Even small amounts grow dramatically over decades (compound interest!)

2. Pay highest-interest debt first

Don't spread payments equally—target the most expensive balance

3. Be aware of card vs. cash

If a purchase feels painless, ask: “Would I pay this in cash?”

4. Use nudges on yourself

Set up automatic transfers to savings—let the default work *for* you

5. Borrow wisely for education

Understand loan terms *before* you sign—student debt follows you

Thank you!

Don't forget to fill out the exit survey.